

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 4th Year 1st Term Examination-2019
Computer Integrated Manufacturing Systems (CIMS)
LE 4123

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

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|------|---|----|
| 1(a) | Is the CIMS application feasible in the footwear industry of Bangladesh? Express your opinion. | 06 |
| 1(b) | Differentiate between solid modeling and surface modeling with necessary figures. | 06 |
| 1(c) | Define Finite Element Analysis (FEA). State the applications of FEA in product design and performance analysis. | 08 |
| 1(d) | Define meshing. Why meshing is crucial before design simulation analysis? | 06 |
| 1(e) | Write short notes on: (i) Von Mises Stress (ii) Buckling Deformation and (iii) Drop test. | 09 |
| 2(a) | Define Computer Integrated Manufacturing (CIM)? "CIM the Enterprise wide solution" – Explain the statement. | 06 |
| 2(b) | "CIM accomplishes each of the major functional area of the manufacturing of a manufacturing enterprise" state in brief. | 10 |
| 2(c) | Draw a typical product life cycle chart for the design process and the manufacturing process used in the CIM system. | 07 |
| 2(d) | Point out the steps of the operational sequences of a design simulation analysis. | 04 |
| 2(e) | How computational model of the footwear can be used to optimize functional performance of the therapeutic, sports or casual footwear for general publics or to fulfill the subject specific requirements? | 08 |
| 3(a) | Define transfer line and stand-alone NC machines. Describe the relative position of the CIMS concept using graphical representation. | 08 |
| 3(b) | Illustrate the relationship among special manufacturing system, manufacturing cell and flexible manufacturing system using a graphical data with their definitions. | 10 |
| 3(c) | What is meant by material handling system? Distinguish between primary and secondary material handling system with their individual requirements. | 09 |
| 3(d) | Define the following CIMS terms:
(i) Traffic control (ii) Part production file (iii) Status report and (iv) System manager. | 08 |
| 4(a) | How to digitize shell lines and style lines using shoe master software? | 05 |
| 4(b) | What is pattern engineering? What are the functions performed during pattern engineering? | 05 |
| 4(c) | What is spring? Why spring is necessary for some piece developments? | 05 |
| 4(d) | Classify types of line modifications in shoe master software. Mention the types of allowances used during piece development including their measurements. | 06 |
| 4(e) | Discuss shortly the sequential process of 2D pattern engineering and grading to develop a completed set of pattern for shoe manufacturing. | 08 |
| 4(f) | State the principle and flow chart of the following shoe designing software:
(i) Cintech Grad system and (ii) Cintech Descom system | 06 |

SECTION-B

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|------|--|----|
| 5(a) | Define automation. Describe the reason for automating. | 12 |
| 5(b) | Describe the different category of production systems and show the relation between production quantity and product variety. | 12 |
| 5(c) | Write notes on (i) Digitizer and (ii) Potter. | 06 |
| 5(d) | What is meant by digitizing? State the objectives of digitizing. | 05 |

- 6(a) What is the basic principle of CAD system for leather product designing? Describe the main parts of CAD. 12
- 6(b) What is Conveyor? "Conveyer is crucial for production line to increase productivity"--Explain the statement. 03
- 6(c) What is numerical control? Classify NC with example. 07
- 6(d) Write down the pros and cons of a CNC machine. 08
- 7(a) Define industrial robot. Classify industrial robot based on different parameters. 08
- 7(b) Describe the following robot with figure: 12
(i) Cylinder configuration robot (ii) Polar configuration robot.
- 7(c) Show the different types of joints commonly used in industrial robot. 10
- 7(d) Discuss briefly about the industrial application of robot specially in footwear manufacturing industry. 05
- 8(a) Given the AGVs layout shown in Fig. 8(a) vehicles travel counter clockwise around the loop to deliver loads from the load station to the unload station. Loading time at the load station is 0.65 min, and unloading time at the unloading station is 0.45 min. It is desired to determine how many vehicles are required satisfying demand for this layout if a total of 30 del./hr. must be completed by the AGVs. The following performance parameters are given: Vehicle velocity = 30 m/min, availability = 0.85, traffic factor = 0.80, operator efficiency = 70%. Determine 20
(i) Ideal delivery cycle time and
(ii) Number of vehicles required to satisfy the delivery demand.

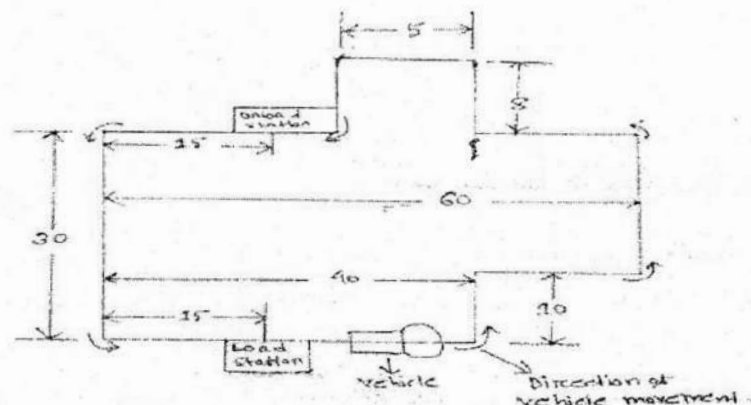


Fig. 8(a) AGV load layout.
Dimensions in meters (m)

- 8(b) What is AGV? Why are AGVs used in industry? 06
- 8(c) What is input device and output device? Mention the name of some input device. 09

Department of Leather Engineering
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 B. Sc. Engineering 4th Year 1st Term Examination-2019
Leather Product Engineering III
LE 4109

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION--A

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|------|--|----|
| 1(a) | Briefly explain the chronological summary of clothing history. | 10 |
| 1(b) | "How far have we got in our attempt to shed light on the reasons people wear cloth?"—Explain briefly. | 15 |
| 1(c) | Mention popular clothing for men's and women's in two distinguish ancient era like Egypt and Gothic. | 07 |
| 1(d) | What is the notion of Anthropologist Malinowski in case of protection? | 03 |
| 2(a) | What is speciality of jacket? Draw a perspective view (front & back) of a classic jacket and identify its different parts. | 10 |
| 2(b) | Define and specify with related figure for the following terminology:
(i) Gorge (ii) Bread line (iii) Seat and (iv) Break point | 12 |
| 2(c) | Why BP and FP are using body block construction of classic Jacket? | 05 |
| 2(d) | Define front opening system. Explain the following two front opening systems:
(i) bottom with button whole and (ii) Stud | 08 |
| 3(a) | Mention the significance of grading and enlargement. | 06 |
| 3(b) | Illustrate the enlargement techniques for Blowson jacket (front panel) with related figure. | 15 |
| 3(c) | Explain the sleeve adaptation techniques in case of easy fitting armhole line:
(i) Lowering the armhole line and (ii) Extended the shoulder | 14 |
| 4(a) | Define pleat and slit. | 04 |
| 4(b) | If the waist girth and the length of a women skirt are 65 cm and 57.5 cm respectively then calculate the waist and length radii accordingly. | 06 |
| 4(c) | What is skirt? How can you give fullness to the skirt block? | 10 |
| 4(d) | Briefly explain the construction process of 12 size straight skirt block. | 15 |

SECTION-B

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|------|--|----|
| 5(a) | Classify waist coat. Mention the speciality of a waist coat. | 09 |
| 5(b) | What is bodice block? Briefly analysis the bodice block of a waist coat. | 14 |
| 5(c) | Describe the following techniques to solve the fitting problems of a waist coat:
(i) Long back (ii) Wide and narrow shoulder (iii) Forward stance | 12 |
| 6(a) | Draw a perspective view of a trouser by introducing all style lines. | 06 |
| 6(b) | Describe the following preparation for a trouser:
(i) Front pocket with extension (ii) Back with yoke and (iii) Zip guard | 12 |
| 6(c) | Mention the necessary measurements required to draft a trouser block. | 05 |
| 6(d) | Describe the trouble shooting for a trouser. | 12 |
| 7(a) | Define fringe. How you compare between frayed fringes and knotted fringes? | 10 |
| 7(b) | If a gentleman has a waist of 35 inch, calculate his approximate chest and half across back. | 08 |
| 7(c) | What are meant by fittings and fastener? Mention some fittings and fasteners for the fabrication of leather goods. | 12 |
| 7(d) | Mention the components of a rivet. | 05 |
| 8(a) | Write short notes on the following: (i) Warp over (ii) Splash suit (iii) Liberty cap (iv) Pelerine | 16 |
| 8(b) | How do accessories differ from fashion accessories? List out major trimmings for a leather jacket. | 15 |
| 8(c) | Cite some distinguish features of frock. | 04 |

Department of Leather Engineering
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B. Sc. Engineering 4th Year 1st Term Examination-2019
LE 4029
Tannery Waste Management

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

- 1(a) Draw a curve for BOD exerted and remaining with respect to time for organic wastewater and derive mathematical relationship between them. 12
- 1(b) Which beamhouse operation has highest pollution load? Justify your opinion. 08
- 1(c) BOD of a sewage incubated for 3 days at 27°C was measured 110 mg/L. Calculate BOD₅ at 20°C. Consider $k=0.23$ per day (base e) and $\theta=1.047$. 08
- 1(d) Write about carbonaceous BOD and nitrogenous BOD. 07
- 2(a) List out the discharge standard of tannery waste water to inland surface according to the environmental conservation rules of Bangladesh (ECR-97). 06
- 2(b) When the concentration of pollution is not very high in a wastewater treatment plant then which coagulation principles is applied? Write about the mechanism of that principle. 14
- 2(c) State the principles of coagulation. Write down the factors affecting coagulation operations. 10
- 2(d) Define sludge volume index (SVI) and surface overflow rate (SOR). 05
- 3(a) Write about the relationship between the dosage of coagulant and coagulation process. 12
- 3(b) Write down the objectives of biological treatment process. 08
- 3(c) A fine bubble aeration system is used to remove BOD and fully nitrify the waste water. The temperature of the waste water is 15°C, site elevation is 5000 ft, the diffuser depth is 18 ft and there is 2.0 mg/L residual dissolve oxygen. Determine the oxygen demand, total horse power requirement and airflow requirement using an influent flow of 2 mgd, 750 mg/L BOD primary effluent, 50 mg/L in the plant effluent, and 15 mg/L ammonia. Dissolved oxygen at 5000 ft elevation 8.8 mg/L, $\alpha=0.65$, $\beta=0.95$ and $F=0.75$ and SOTR for the bubble diffusers is 4.0-7.7 lb hp-hr. 15
- 4(a) Write down the design criteria of aeration tank for conventional activated sludge process. 07
- 4(b) What are the differences between slow sand filter and rapid sand filter? 03
- 4(c) How granular media filter works during treatment? Write the mechanism of filter with proper drawing. 10
- 4(d) Define RO. Write the principle of RO. Also state the advantages of RO technology over trickling filter. 10

SECTION-B

- 5(a) Write down the toxicity of heavy metal components present in the tannery solid waste. 08
- 5(b) Mention the objectives and strategies of solid waste management (SWM) 07
- 5(c) State the environmental impacts of tannery solid waste. 08
- 5(d) Discuss the characteristics and composition of solid waste generated from tanning process. 12
- 6(a) How can utilize the Cr shaving/splitting as gelatin and poultry feed? 08
- 6(b) State the environmental model of a safe disposal of tannery solid waste as landfill with proper figure. 10
- 6(c) Briefly discuss the process of waste water (WW) treatment for tannery waste. 08
- 6(d) Describe the utilization of- i) Finishing trimming ii) Buffing dust iii) wet white trimming & shaving. 09
- 7(a) How can remove the sulphide from tannery solid waste-Explain. 10
- 7(b) Mention the standard values of physicochemical parameters of composting for tannery solid waste. 06
- 7(c) What is waste to energy (W to E)? Explain the technology of biogas production from tannery solid waste. 10
- 7(d) Mention the name of different process and technology to change the waste to energy (WTE). Explain the biological conversion process. 09
- 8(a) Define sustainable development. Write down the components of sustainable development in tanning industry. 08
- 8(b) What is cleaner technology? How can reduce water consumption in tanning industry. 07
- 8(c) How can you recover the salt from curing and used lime liquor for further uses? 08
- 8(d) Discuss the separation process of chromium from tanning effluent by polyelectrolytes with figure and parameters. 12

Department of Leather Engineering
 Khulna University of Engineering & Technology
 B. Sc. Engineering 4th Year 1st Term Examination-2019
 Footwear Engineering III
 LE 4111

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

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|---|----|
| 1(a) Define 1 st waste, 2 nd waste and 3 rd waste based on Russ and Small method. | 06 |
| 1(b) Describe briefly about the marking up method for material allowance calculation with its advantages and disadvantages. | 10 |
| 1(c) State the features of different Swedish strip steel knives that are used for cutting shoe materials. | 09 |
| 1(d) Write short notes on the following die-less cutting systems:
(i) Laser cutting and (ii) Water-jet cutting | 10 |
| 2(a) Mention the top ten golden rules for closing shoe uppers. | 10 |
| 2(b) Write down the process of developing 'U'-bound topline with figures. Which points are required to maintain during making this topline? | 09 |
| 2(c) Write short notes on the following shoe top lines: (i) Collar topline and (ii) Bagged topline | 10 |
| 2(d) What are the possible causes and their suggested remedies for overhanging sections after shoe upper sewing operation? | 06 |
| 3(a) State some ideal characteristics of shoe topline tape. | 05 |
| 3(b) List out the occupational health hazards in footwear industry. | 05 |
| 3(c) What are the functions of the shoe specification sheet? What does the header of shoe spec sheet contain? | 08 |
| 3(d) Define plant layout. Write down the advantages of a good plant layout. | 08 |
| 3(e) Briefly discuss the problems of footwear that may occur during long-time storage due to improper conditioning. | 09 |
| 4(a) Mention the features of different shoe grades in case of quality inspection. | 07 |
| 4(b) Sketch a shoe and indicate common shoe problems that are found in quality inspection. | 09 |
| 4(c) State the basic principles of shoe costing. | 07 |
| 4(d) Write short notes on the following costing terms with example:
(i) Depreciation cost (ii) Royalty cost and (iii) Semi-variable cost | 12 |

SECTION-B

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|--|----|
| 5(a) "Insole is the most important material for shoe production"- Justify the statement. | 05 |
| 5(b) Write down the complete operational sequences in the lasting department. | 15 |
| 5(c) How insole beveling operations ensure the smooth feather edge of footwear? | 05 |
| 5(d) Point out the prime factors for successful insole moulding operation. | 10 |
| 6(a) Explain the basic points to look for a successful seat and side lasting operation. | 05 |
| 6(b) Write short notes on: (i) Priming and (ii) Solvent vapour degreasing (SVD) | 08 |
| 6(c) State the possible cause and remedy of the following problems of seat and side lasting operation: (i) Not enough adhesive (ii) Adhesive not being extended in the correct position. | 06 |
| 6(d) Explain the stitching systems of the toe and forepart lasting machine. | 12 |
| 6(e) Write shortly the importance of temporary attachment in the lasting room. | 04 |
| 7(a) Explain about the combined Californian and cemented shoe construction process. | 15 |
| 7(b) State the PVC injection moulding construction with bottom preparation. | 12 |
| 7(c) List out the major pros and cons of PU injection shoe construction. | 08 |
| 8(a) "Heat is crucial for crease and wrinkle removal" – Justify the statement. | 07 |
| 8(b) Define wrinkle chasing. Describe briefly about the different ways of wrinkle chasing from shoe upper. | 08 |
| 8(c) State about the repairing agents with their main features. | 10 |
| 8(d) There are lots of operations for crease removal from the shoe surface. Which process is more convenient from your point of view? | 10 |

Department of Leather Engineering
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B.Sc. Engineering 4th Year 1st Term Examination-2019
LE 4121

Environmental Science & Engineering

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

- 1(a) How the ecological destruction led to damages in economy and human health? Discuss with example. 10
 1(b) Explain the structure of an ecosystem. 08
 1(c) What is tropic level? Differentiate between food chain and food web 09
 1(d) Define climate change and global warming-how are they related? 08
 2(a) State the significance of wet lands in the context of Bangladeshi environmental conditions. 12
 2(b) Differentiate between tropospheric ozone and stratospheric ozone. 08
 2(c) Discuss briefly sources and effect of the following air pollutant(s): i) Smog ii) Particulate matter iii) Nitrogen dioxide 12
 2(d) Define water pollution. 03
 3(a) Write down the sources, standard values and significance of the following water quality parameters: i) Alkalinity ii) Iron, iii) TDS and iv) BOD 12
 3(b) What eutrophication? How can it be controlled? 06
 3(c) Write down the sources of ground water pollution. 06
 3(d) A water has the following analysis: 11

Ion	Unit (mg/L)	Ion	Unit (mg/L)
Na ⁺	20	Cl ⁻	40
K ⁺	45	HCO ₃ ⁻	60
Ca ²⁺	7	CO ₃ ²⁻	1
Mg ²⁺	9	SO ₄ ²⁻	7
Sr ²⁺	3	NO ₃ ⁻	8

Find the total hardness, carbonate hardness and noncarbonated hardness.

- 4(a) Write down the effect of hard water on the following leather processing operation: i) Vegetable tanning ii) Dyeing 10
 4(b) Differentiate between point sources and nonpoint sources of water pollution. 06
 4(c) What is indicator organism? Write down the characteristics of indicator organism 06
 4(d) What size of sample is required if the 5-day BOD is 300 mg/L and the total oxygen consumed in the BOD bottle is limited to 4 mg/L? 12

SECTION-B

- 5(a) How ecotoxicology, ecology, toxicology and toxic substances are interlinked with each other. 05
 5(b) State at least five toxic chemical agents used in leather manufacturing including their uses and human organ at risk. 09
 5(c) Concisely illustrate the attacking phenomena of toxicants. 12
 5(d) Describe briefly the COSHH guideline for the control of hazardous substance to health. 09
 6(a) What is meant by the conservation of environment? Write down the procedure for issuing environmental clearance for "Red" category projects. 11
 6(b) What is said by section 5 of ECA about the declaration of ecologically critical area? What are the prominent factors that should be considered by the government to declare any area as ecologically critical area? 10
 6(c) Mention the discharge standards set by the ECR for waste from tannery industry. 05
 6(d) Write down the short note of the following terms: i) enforcement ii) compliance monitoring and iii) auditing 09
 7(a) What IEE? Write down the purposes of EIA. 06
 7(b) Discuss briefly the EIA process with flow diagram. 10
 7(c) How baseline study depends on scoping? Write down the part and parcel of ToR. 10
 7(d) Write down the advantages and disadvantages of matrices, networks, and overlay methods of impact identification. 09
 8(a) When the Ad-hoc method can be effective? Concisely illustrate the ad-hoc method for comparing alternative industrial units or projects arrangements. 12
 8(b) Write down the steps involved in obtaining ECC for a tannery industry. 09

- 8(c) Tannery industry will be located on the south bank of the river Ruphsa. The width of the river at the plant site is about half kilometer. There is a forest and some paddy field spread over on the east side of the plant. Two alternative plant proposals with predicted impacts submitted to the DoE. The environmental impacts of the plant will be assessed using EES. Values of the environmental qualities with and without project and environment importance of difference parameters are given below:

Parameters	Values of environmental qualities			Environmental importance
	Without project	With project for alternative 1	With project for alternative 2	
Forest area	0.8	0.2	0.4	65
Air quality	0.8	0.3	0.3	88
Land value	0.2	0.6	0.5	64
Water quality	0.6	0.2	0.4	152
Agriculture	0.6	0.3	0.2	60
Human health	0.8	0.4	0.5	95
Socio-economic condition	0.2	0.9	0.8	120
Life patterns	0.3	0.6	0.8	55
Ecology	0.3	0.8	0.7	148
Aesthetic	0.7	0.3	0.2	-

Find the environmental importance for aesthetics of the plant area and also determine which plant proposal will be environmentally more sustainable.